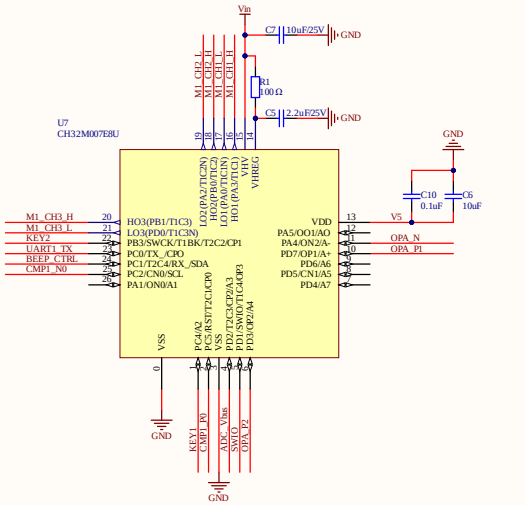
MCU



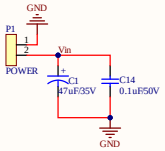
When laying out pins, watch out for keeping high-speed and high-current pins away from sensitive sampling lines

If UART1_TX and the like are high-speed signal lines when hooked up to a virtual oscilloscope

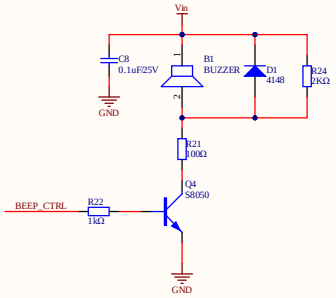
Try to avoid running parallel to the sinulated sampling signal, and it's recommended to add an RC to the serial port to reduce edge rise/fall rates.

The Vhv limit can't go over 26V

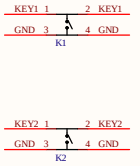
POWER



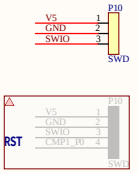
BUZZER



KEY



SWD



This series of MCU has a single-wire programming feature

When the reset function is enabled on PC5, to avoid the inability to burn,

It is recommended to lead out this pin to the SWD interface.

Reset via LINK during programming

Phase current sampling

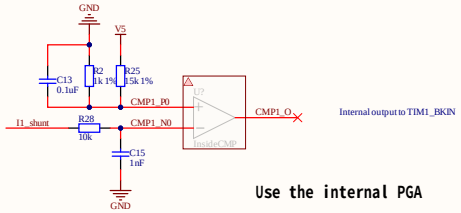


Use the internal PGA

UART

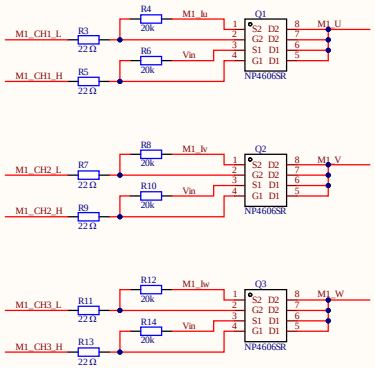


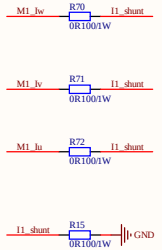
Comparator overcurrent

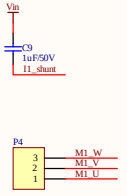


Use the internal PGA

Inverter circuit







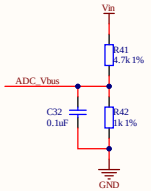
Select the Kelvin connection wiring method

Pseudo-differential routing

Select the resistance and package according to the actual current

Adjust the gate drive resistance value according to the actual conditions such as trace length and thickness, with the purpose of reducing gate oscillation

Bus voltage detection



Adjust the resistance and package of each resistor according to the value of Vin

Title		
Size: C	Number	Revision
Date: 11/05/2026		Sheet of
File: F:\CH32M007E8U_FAN_R0.SchDoc		Drawn By:

Comment	Description	Designator	Footprint	LibRef	Quantity